

ChBE 4803/8803 AI for Experimental Chemical Engineers (and Beyond!)

Fall 2026

Location: ES&T L1175

Time: 8-9:15 am

Credits: 03

Prerequisites:

Background in chemical engineering and Python programming

Pre- and/or Co-Requisites:

- 8803: Graduate students are encouraged to take ChBE 6745 and ChBE 6746 prior to taking this course. Students should be familiar with basic concepts in math, linear algebra, numerical methods, optimization, python programming, and data analytics to be successful in this course.
- 4803: Prerequisites are: CHBE 3215 OR BMED 3310 OR MSE 3210

For both undergraduate and graduate students taking this course, you do not need to have taken any Python course before, but be aware that we will use Python codes in this course. If you already know how to program in Python, great; if not, you will need to invest time to become literate in Python (all your knowledge from MATLAB can be easily transferred to Python). **Module 0** (See Files/Resources) of this course will offer resources that you can use to improve your Python programming.

Instructor:

Professor Vida Jamali

vida@gatech.edu

Office hours: Tuesdays 9:15-10:15 am ES&T 1229

Teaching Assistants:

Dimitrios Fardis

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Office hours: TBD

Course Description: This course has two primary objectives: (a) to introduce students to advanced deep learning and AI methods and (b) to teach students how to use such models on experimental data relevant to chemical engineering and other disciplines, including biomedical engineering, materials science and engineering, and chemistry. Chemical (as well as bio and materials) engineers work with experimental data of various modalities that span microscopy images, time series data, spectrophotometry data, and texts, to name a few. This course will first introduce students to the foundations of popular AI models developed in the last 10 years and their application in analyzing experimental data. In addition to covering various model architectures (e.g., CNN, VAE, LLM, agentic AI, etc.), we will cover foundational topics related to AI, including framing loss function, initialization, error backpropagation, gradient computation, and model fitting. The course will maintain a balance between neural network basics and application to experimental data at hand, such that the students obtain key knowledge on how to efficiently select the most suitable AI model and apply it to the data, know the mathematical principles of the model operation,

test the model performance, and relevant statistical analysis. Overall, by the end of the course, the students will have a good understanding of how to apply AI models to experimental data relevant to chemical engineers and extract useful information.

Specific Outcome: At the end of this course, the student will:

- have a good understanding of how AI models operate: Students will be able to explain the principles and operation of popular AI models (e.g., Convolutional Neural Networks (CNN), Variational Autoencoders (VAE), and Large Language Models (LLM)) and understand their relevance to analyzing experimental data,
- be able to apply AI models to experimental data: Gain hands-on experience in applying AI models to various types of experimental data common in chemical engineering, including microscopy spectroscopy, synthesis, biomolecular engineering, and electrochemistry,
- be able to select a model and implement it: Students will be able to assess which type of AI model is suitable for their data,
- have a mathematical understanding of the AI models: This includes loss functions, optimization, and regularization, and implement the appropriate algorithms accordingly,
- be able to evaluate the performance of the model: Students will learn the relevant tests and statistical methods to evaluate the performance of the AI model and how to report the results,
- be able to communicate their results and present them to their peers.

Recommended textbook and other resources:

Simon Prince, Understanding Deep Learning ([available for free](#) or purchase)

Useful resource:

Terence Parr and Jeremy Howard, [The Matrix Calculus You Need for Deep Learning](#), arXiv

Justin Johnson, [Python NumPy tutorial](#)

[Basic Linear algebra review](#)

Zico Kolter, [Linear algebra review and reference](#)

Class website: <https://gatech.instructure.com/courses/544790>

Web: Course material, homework assignments, screencasts, announcements and grades will be posted on Canvas (<https://canvas.gatech.edu/>). Materials needed for each week will be organized in the Files section. All announcements for this course will be posted on Canvas.

Attendance: Attendance is encouraged. Material presented in class will appear on homework. You are responsible for obtaining class materials if absent for non-health or institute-approved reasons. A Zoom recording of the lectures will be available after the class. However, you should be present on days with in-class activities (paper discussion, project pitch, exams, etc.). In the case of an absence when an exam is given, make-up will only be given when the absence is caused by an official Georgia Tech event or illness; such absences must be documented with a letter from the appropriate Georgia Tech official, a physician or the Dean of Students. **Everyone should feel comfortable asking questions in our class, and should respect the questions of their classmates.**

Piazza: (accessible from Canvas plug-in) <https://piazza.com/class/msqmdvcuoyx4vm>

Piazza is the main communication channel in this course. Emails are strongly discouraged. You will be getting an email from Piazza (to your official GT email) inviting you to join the class Piazza site. We will be conducting all class-related discussions here this semester. The quicker you begin asking questions on Piazza, the quicker you'll benefit from the collective knowledge of your classmates and instructors. We encourage you to ask questions when you're struggling to understand a concept or a homework problem or just have a general question about class procedures—you can even do so anonymously. Here are the ground rules:

1. TAs and the instructor will only respond to questions 24 hours after they are posted. This is to encourage you to help and answer each other -- part of what Piazza is all about!
2. We will "endorse" answers from students as soon as possible so you know that the answers your classmates posted are "correct".
3. We will not answer any questions via email, except those of a personal nature. Please use Piazza so everyone can benefit.
4. Top 3 responders (with at least 3 responses) will receive 1 bonus point.

Grading Scheme:

(The final grade cutoffs are subject to change)

Homework	30% (for undergrads 40%)	A	90-100 %
Midterm I Exam	10%	B	80-90 %
Midterm II Exam	10%	C	70-80 %
Student Paper Presentation	10%	D	60-70 %
Project	40% (for undergrads 30%)	F	0-60 %

Homework:

Homework (electronic copy uploaded on Canvas unless the problem statement specifies otherwise) is due at the beginning of class (8:00 am). Assignments will be considered late if uploaded after 8:05 am. Unfortunately, we won't be able to accept late HWs to be fair and so solutions can be posted promptly. Everyone is entitled to a one-time, 1-day (business day) extension (i.e., homework due Thursday would be due by 8:00 am Friday). No need to email me to indicate the reason. a) This is separate from any other accommodation from the Office of Disability Services, and b) This is only valid for HWs, not the project.

Solutions will be posted on Canvas ~at the end of class (or next business day if late submitters). Students are responsible for determining their mistakes from the solutions or office hours. Working with others is allowed and encouraged to help each other learn, but each student must write up and turn in their own work (see Honor Code). Homework copied from classmates, former students, solutions from previous semesters, solution manuals, or LLMs is considered cheating and will be reported.

Homework Grading:

All problems are worth 3 points.

0 = didn't try it at all

1 = tried some of the problem, or tried most of it but did it mostly wrong

2 = tried most or all of the problem but had some errors

2.5 = tried the whole problem and it was nearly correct with very minor error

3 = tried the whole problem and it was entirely correct

Project: Semester-long project for developing an AI model for experimental data sets coming from a lab at GT. The project will be completed in teams. You must submit the project proposal by Jan 30th. Collaboration and teamwork are essential components of these projects, and professional behavior and respect for your teammates are critical. You and your teammates will pick a project topic and discuss it with the instructor during office hours in September to ensure that it is suitable for this course. The final project will be graded based on the following breakdown. More details/guidelines will be provided for each assignment and are subject to change. Detailed rubrics and deadlines are posted on Canvas. Assignments in **bold will be instructor-graded, and non-bold assignments will be peer-graded.**

- 10% - **Project proposal** (1 file per team): ~1-page proposal on the description of dataset and project goals
- 10% - Data preparation: Dataset ready for analysis (no missing values, etc.)
- 10% - **Baseline model:** Simple model evaluated with an appropriate assessment metric.
- 15% - Improved model: More accurate or efficient model with assessments.
- 10% - **Project pitch:** In class project pitch for 5 minutes with slides including some preliminary result.
- 10% - **Report draft:** Draft of ~5 page project report detailing approach and results.
- 15% - **Final Report:** Finalized version of the project report.
- 5% - Work ethics and team work (graded by teammates):
- 15% - Final Poster Presentation: Last day of class.

Note: Some parts of the project will be due before a topic is covered in the course (e.g., data preparation). You may revise your strategies after we cover a topic in the course but should be able to establish a simple strategy before a topic is covered by attending office hours and looking at course notes ahead of time.

Faculty-led student paper presentations:

There will be **four** student paper presentations and faculty-led discussions with days indicated on the syllabus (two papers each time, 8 papers total). Each paper will be assigned a defense and offense team. The defense team will defend the paper for 15 min (as if this is their own research paper), and then the offense team will attack the paper for 15 min, followed by 5 min Q&A. Each team can have up to two members. We will have a five-minute break before starting the discussion on the second paper of that day. The papers preselected (by the instructor) cover various topics of chemical engineering research in which using AI (AI models covered in lectures) resulted in significant advancement in that research area. The papers will also introduce new directions that could be useful for the final project. **All students in the class must sign up by September 1st to present one of the papers. Please use this link to sign up:**

[PaperDiscussion-Team-SignUp.xlsx](#)

Lecture Notes:

Slides for the course are presented in PowerPoint/Keynote and will be shared with students on the course Canvas page. Lecture notes will be continuously updated throughout the course so please check back regularly.

Honor Code:

Students are expected to follow the Georgia Tech Honor Code at all times.

<http://www.honor.gatech.edu>

Copying from other students (current or former), instructor/TA solutions from previous semesters, solution manuals, websites, AI-generated solutions, pay-for-work services, or any other form of the word is considered an honor code violation and will be reported along with receiving a zero on the assignment.

Posting the HWs and the solution to HWs on Github (or any other public repository) is a violation of the honor code.

Disability Statement: My goal is to create a learning environment that is welcoming to everyone in the class. Please be kind, respectful and professional both in and out of our classroom. If there are aspects of the instruction or design of this course that result in barriers to your accurate assessment, or achievement, please notify me as soon as possible. Students with disabilities should contact the Office of Disability Services to discuss options for removing barriers in this course, including accommodations. ODS can be reached at 404-894-2563, dsinfo@gatech.edu, or disabilityservices.gatech.edu. If you need accommodations for exams, please contact the ODS office as soon as possible and make reservations for the days based on the schedule below for all exam dates.

Institute-Approved Absences: As per Georgia Tech policy, you are permitted to be absent from class to participate in athletic events, official field trips, and religious observances. For planning purposes, please provide me with written notice of your upcoming absence at least two weeks before the event, and ideally within the first two weeks of class. When I receive this notice, you and I will discuss opportunities to make up work you will miss in your absence. Please see <http://catalog.gatech.edu/rules/4/> for more information about receiving official notice from the Registrar about the nature and timing of your upcoming Institute-approved absence.

Campus Resources for Students: If physical or mental health, hunger, or homelessness are affecting your studies, please reach out to one of these resources or to me.

- [Counseling Center](#)
 - Crisis Line: 404.894.2575 (M-F 8 a.m.- 5 p.m.)
 - After-hours counselors: 404.894.2204
- [Dean of Students and Student Life](#) (404.894.2565)
- [Georgia Tech Police Department](#) (404.894.2500)
- [Belonging & Student Support Center](#)
- [Veterans Resource Center](#)

Tentative Class Schedule

Module 0

Resources for Python Programming:

<https://cs231n.github.io/python-numpy-tutorial/>

Matplotlib tutorial:

<https://github.com/rougier/matplotlib-tutorial>

Resources for reviewing linear algebra, vector calculus, and probability

<https://stanford-cs221.github.io/spring2023/modules/index.html>

Resources for reviewing linear algebra

[Linear Algebra Review by Zico Kolter](#)

Week	Lec #	Date	Lecture Title	Due	
1	1	8/25/26	Introduction- what is AI?		T
1	2	8/27/26	Supervised vs. unsupervised models		Th
2	3	9/1/26	Linear models to generalized linear models and their application in thermodynamic property prediction	Paper presentation-signup, HW 1	T
2	4	9/3/26	K-nearest neighbor, Linear classifiers for nanoparticle classification		Th
3	5	9/10/26	Deep Neural Networks (DNNs): Loss function		Th
3	6	9/15/26	DNN: Regularization and optimization		T
4	7	9/17/26	DNN: Backpropagation	HW 2	Th
4	8	9/22/26	Training a neural network-part I	Project proposal	T
5	9	9/24/26	Training a neural network-part II		Th
5	10	9/29/26	Guest Lecture: Prof. Solomon Oyakhire, GT		T
6	11	10/1/26	Midterm I		Th
6		10/6/26	Fall Break		T
7	12	10/8/26	Training a neural network-part II.2, Convolutional Neural Networks (CNNs)		Th
7	13	10/13/26	CNN	HW 3	T
8	14	10/15/26	Dilated CNNs, Application of CNN in particle tracking and time series data analysis		Th

8	15	10/20/26	Paper discussion: 1) Application of CNNs in Raman spectroscopy data analysis 2) self-supervised denoising for electron microscopy 3) ML for discovery of polymer gas separation membranes	Baseline model due	T
9	16	10/22/26	Unsupervised models (AEs), UNet architecture		Th
9	17	10/27/26	Paper discussion: 1) UNet for cell microscopy image analysis 2) deep generative priors for electron ptychography	HW 4	T
10	18	10/29/26	Project pitch with some prelim result	Project Pitch due	Th
10	19	11/3/26	Generative AI, Variational Auto Encoders (VAEs)		T
11	20	11/5/26	From Brownian motion to Diffusion models	HW 5	Th
11	21	11/10/26	Guest Lecture: Dr. Cory Hargus, Entalpic		T
12	22	11/12/26	Paper discussion: Diffusion models for 1) antimicrobial peptide design ; 2) representation learning of mass spectrometry data		Th
12	23	11/17/26	Midterm II		T
13	24	11/19/26	Large Language models	HW 6	Th
13	25	11/24/26	Agentic AI		T
		11/26/26	Thanksgiving Break		Th
14	26	12/1/26	Automating Chemical labs	Project report draft	T
14	27	12/3/26	Paper discussion: Automating chemical labs in practice: 1) agentic self-driving lab for protein function prediction ; 2) Self-driving labs for catalysis		Th
15	28	12/8/26	Poster Presentation (ES&T L1 atrium)	Final project report due on 8th	T